

Practice Questions for Annuity (Present value, Future value, Perpetuity, Effective annual rate)

Question 1

Find the future values of these ordinary annuities. Compounding occurs once a year.

- a. \$500 per year for 8 years at 14%
- b. \$250 per year for 4 years at 7%
- c. Rework parts a, b assuming they are annuities due.

Question 2

Find the present values of these ordinary annuities. Discounting occurs once a year

- a. \$600 per year for 12 years at 8%
- b. \$300 per year for 6 years at 4%
- d. Rework parts a, b assuming they are annuities due.

Question 3

What is the present value of a \$600 perpetuity if the interest rate is 5%? If interest rates doubled to 10%, what would its present value be?

Question 4 (if covered)

Bank A pays 2% interest compounded annually on deposits, while Bank B pays 1.75% compounded daily.

- a. Based on the EAR (or EFF%), which bank should you use?
- b. Could your choice of banks be influenced by the fact that you might want to withdraw your funds during the year as opposed to at the end of the year? Assume that your funds must be left on deposit during an entire compounding period in order to receive any interest.